The Case for Voluntary Disclosure: Strategic Feature Revelation in Regression and Classification

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Abstract

This paper addresses a fundamental design question: should a predictive model enforce mandatory feature disclosure or allow voluntary disclosure by participants? Since participants strategically reveal only those features that align with their private objectives, the model must account for opportunistic reporting. For the regression problem, we establish conditions under which the predictive model actually achieves better performance by permitting selective revelation. We extend these results to the classification setting. Finally, we explore a dynamic learning framework where the model interacts repeatedly with participants, presenting an algorithm that adaptively switches between disclosure regimes with established theoretical performance bounds.

1 INTRODUCTION

The performance of any predictive model depends on the set of features provided for prediction. In standard settings, all relevant features are assumed to be fully observed by the model. In this paper, we study a setting in which a participant can voluntarily choose which features to disclose. It is natural to assume that the participant will disclose only those features that are beneficial to their personal objective. The central question we address is: under what conditions does voluntary feature disclosure actually improve predictive performance?

For a predictive model to benefit from such disclosure, its inference procedure must account for the possibility of opportunistic reporting. This problem becomes intricate when participants do not share a uniform directional preference over outcomes, as these heterogeneous preferences make it difficult to exploit systematic patterns in feature withholding.

A concrete and illustrative example arises in financial markets when estimating a firm's valuation. The market may enforce a mandatory disclosure rule for financial information. If full disclosure is required, the resulting valuation achieves a baseline accuracy dependent on the complexity of the underlying relationship. Alternatively, the market might allow firms to voluntarily disclose financial details, in which case the valuation is formed based only on the revealed information.

Firms typically hold privately preferred valuations that differ from the market's assessment, and the preferred direction of this deviation is not uniform across firms. Some firms actively seek overvaluation because it raises stock prices and lowers the cost of equity financing. Others may prefer undervaluation, for example, when preparing for a stock buyback program or seeking to avoid triggering regulatory scrutiny.

If a firm's preferred valuation is closely aligned with the market's objective, selective feature disclosure may assist by intuitively filtering out noisy or less informative signals, thereby enhancing valuation accuracy relative to mandatory disclosure. In contrast, if the firm's preferences are misaligned with the market's objective, selective disclosure distorts the available information and introduces a systematic bias, thereby degrading the valuation accuracy.

In the speical case if firms shared the same directional preference—for instance, if every firm preferred overvaluation—the market could prevent exploitation simply by assigning a conservatively low default valuation in cases of partial disclosure. Such a rule would effectively discourage withholding, thereby replicating the outcome of a mandatory disclosure regime. In this work, however, we accommodate the more general and realistic setting in which participants hold heterogeneous preferences. We derive conditions to identify when the strategic disclosure of features is beneficial to the predictive model.

1.1 Main Contributions

In our work, we consider the predictive problems of classification and regression, modeling the strategic interaction as a sender-receiver game.

Firstly, assuming the environment is fully known to both players, we establish sufficient conditions in the regression setting for the predictive model to benefit from voluntary disclosure. These conditions depend on the misalignment between the sender's preferred value and the ground truth value. We also provide necessary conditions that depend on the relationship between the disclosed features and the ground truth.

Secondly, we extend our analysis to the classification problem, establishing necessary and sufficient conditions for strategic advantage in terms of the structure of various conditional distributions.

Thirdly, we explore a learning version of the problem where the sender and receiver interact repeatedly. In this dynamic setting, the receiver seeks to minimize their cumulative loss and can actively switch between voluntary and mandatory disclosure regimes. We present a learning algorithm for this setup and establish theoretical bounds on its performance.

1.2 Related Works

Our work builds on the intersection of strategic machine learning, sender-receiver games, and voluntary disclosure in predictive modeling. Early studies in strategic classification, such as [1], formalized settings where agents manipulate features to influence outcomes, showing that robust classifiers can mitigate gaming in linear models. This was extended in [2] to dark (unknown) environments, where agents' responses are not fully observable, leading to equilibrium analyses under uncertainty.

Sender-receiver games provide a foundational framework for our analysis. [3] examined learning dynamics in these games using experimental data, highlighting how repeated interactions evolve into efficient signaling. Dynamic variants, like those in [4], incorporate Markov chains for state evolution, akin to our noisy channel model. Evolutionary perspectives, as in [5], explore multi-agent signaling, while [6] compared stimulus-response and belief-based learning, informing our online regret bounds.

Voluntary disclosure literature addresses selective revelation. [7] modeled evolving news in dynamic settings, deriving conditions for disclosure thresholds. [8] analyzed investor disclosures, predicting good-news bias, which parallels our heterogeneous preferences. Empirical studies, such as [9], used XGBoost and XAI for predicting disclosure patterns, emphasizing transparency in strategic contexts.

Stackelberg equilibria are central to our game-theoretic approach. [10] characterized equilibria in strategic classification and regression, designing incentives to prevent cheating. [11] generalized PAC-learning for heterogeneous agents, bridging classification and regression. In adversarial prediction, [12] framed problems as Stackelberg games, solving for optimal decision rules.

Online learning with strategic agents extends our dynamic framework. [13] studied allocation under strategic behavior, deriving regret-minimizing algorithms. [14] proposed online principal-agent

models for unknown types, achieving sublinear regret. Decentralized settings in [15] model coalition formation among selfish agents, aligning with our adaptive disclosure policy.

While these works advance strategic aspects in classification [16] and regression [17], they often assume known environments or linear models. Our contributions provide formal sufficiency and necessity conditions under noisy channels and heterogeneous preferences, with theoretical bounds for online adaptation, filling a gap in generalized predictive settings.

2 PROBLEM FORMULATION

Let $\mathcal{X} = \mathbb{R}^d$ be the feature space and $\mathcal{Y} = \mathbb{R}$ (or $\mathcal{Y} = \{-1, 1\}$ for classification) be the outcome space. We consider a data-generating process where Nature selects $(u, x, y) \sim (U, X, Y)$ over $\mathcal{Y} \times \mathcal{X} \times \mathcal{Y}$. Here, y is the ground truth value, x represents the available features, and u is the preferred value.

The problem is modeled as a game between two players: a **Sender** and a **Receiver**. The Sender observes (u, x) and chooses an action A from the strategy space:

$$\mathcal{S} = \{A : \mathcal{Y} \times \mathcal{X} \mapsto \{\emptyset, x\}\}.$$

This action represents the Sender's decision to either disclose the full feature vector x or withhold it (denoting the latter by $\emptyset$). The chosen action $A(u, x)$ is transmitted through a noisy channel, resulting in an observed message M. The channel noise is characterized by a parameter $q \in [0, 1]$ such that:

$$M = \begin{cases} \emptyset & \text{if } A(u,x) = \emptyset, \\ \emptyset & \text{if } A(u,x) = x \text{ with probability } q, \\ x & \text{if } A(u,x) = x \text{ with probability } 1-q. \end{cases}$$

Upon observing $M \in \{\emptyset, x\}$, the Receiver applies a strategy $\hat{y} \in \mathcal{R}$:

$$\mathcal{R} = \{\hat{y} : \{\emptyset, \mathcal{X}\} \mapsto \mathcal{Y}\}.$$

The loss of each player is defined as follows: the Sender incurs a loss $\ell(u, \hat{y}(M))$ based on their preferred value u, while the Receiver incurs a loss $\ell(y, \hat{y}(M))$ based on the ground truth value y.

We analyze this interaction as a **screening game**. In this game, the Receiver first commits to a strategy $\hat{y} \in \mathcal{R}$. Then, the Sender observes this commitment and takes an optimal action $A \in \mathcal{S}$ based on $\hat{y}$. Finally, the Receiver executes their committed strategy and the outcome is realized.

For any commitment $\hat{y}$ by the Receiver, the Sender's **best response** is to choose an action $A(u, x)$ that minimizes their own expected loss. This reduces to a pointwise minimization for each observed state (u, x):

$$A(u, x) = \arg\min_{a \in \{\emptyset, x\}} \mathbb{E}\left[\ell(u, \hat{y}(M)) \mid A(u, x) = a\right]. \tag{1}$$

Anticipating the Sender's best response, the Receiver must now optimize the choice of $\hat{y}$ to minimize their own total expected loss, achieving the strategic risk R_S:

$$R_S := \min_{\hat{y} \in \mathcal{R}} \mathbb{E}[\ell(Y, \hat{y}(M))]. \tag{2}$$

Such a pair of strategies $(A, \hat{y})$ constitutes a **Stackelberg equilibrium**.

To evaluate the strategic gain, we compare this equilibrium to a mandatory disclosure regime (we also refer to this as the **vanilla** setting). Under mandatory disclosure, the Sender is forced to reveal features ($A \equiv x$), which leads to the Receiver's optimal strategy $\hat{y}_V$. The corresponding vanilla Risk is:

$$R_V := \mathbb{E}[\ell(Y, \hat{y}_V(M))]. \tag{3}$$

Definition 1 (Strategic Advantage)**.** *A **strategic advantage** exists if permitting voluntary disclosure yields a strictly lower expected loss for the Receiver than the mandatory regime, i.e.,* $R_S < R_V$.

3 MAIN RESULTS

In this section, we characterize the conditions under which a strategic advantage arises across different learning settings. Throughout our analysis, we adopt the shorthand notation $\hat{y}(\emptyset)$ to denote the Receiver's prediction when features are withheld (also called default prediction), $\hat{y}(x)$ when features are revealed, $\hat{y}(M)$ when evaluating on the observed message M, and $\hat{y}_V(M)$ for the vanilla predictor's output on M.

Firstly, we analyze the general regression problem under the assumption of a known environment, deriving distinct necessary and sufficient conditions for the emergence of a strategic advantage. Secondly, we turn to the binary classification problem—also assuming a known environment—and establish the necessary and sufficient conditions for strategic advantage; notably, for classification, these conditions are found to be equivalent. Thirdly, we consider a dynamic setting where the environment is unknown, and the Receiver must interact repeatedly with the Sender. In this case, we propose an online learning algorithm that adaptively selects the appropriate disclosure regime and provide theoretical performance guarantees for the receiver's learning process. Detailed proofs for all results in this section are deferred to the Postponed Proofs section 6.

3.1 Strategic Advantage Condition in Regression

The outcome space $\mathcal{Y} = \mathbb{R}$ is the space of real numbers and the loss function $\ell(y, \hat{y}(M)) = (y - \hat{y}(M))^2$ is mean squared error (MSE). For this regression formulation, we explicitly assume that all random variables (U, X, Y) have zero mean everywhere. In the mandatory disclosure setup, the Sender always chooses $A = x$. For a linear Receiver strategy $\hat{y}_V(\emptyset) = 0$ and $\hat{y}_V(x) = b^\top x$, we characterize the resulting risk in the following lemma.

Lemma 1. *In the mandatory disclosure setup, for a Receiver strategy parameterized by* $b \in \mathbb{R}^d$*, the expected risk is*

$$\begin{aligned} R_V(b, q) = \operatorname{Var}(Y) &- (1-q)\operatorname{Var}(b_Y^\top X) \\ &+ (1-q)\operatorname{Var}((b - b_Y)^\top X), \end{aligned} \tag{4}$$

where $b_Y = \operatorname{Var}(X)^{-1}\operatorname{Cov}(X, Y)$*. The optimal vanilla risk is achieved at* $b = b_Y$*, yielding*

$$R_V^\star(q) = \operatorname{Var}(Y) - (1-q)\operatorname{Var}(b_Y^\top X). \tag{5}$$

Remark 1. *Note that* $b_Y = \operatorname{Var}(X)^{-1}\operatorname{Cov}(X, Y)$ *is the optimal weight vector for ordinary least squares (OLS) regression of* Y *on* X*.*

In the voluntary disclosure setup, for a given Receiver strategy $\hat{y}(M)$, we consider the linear form $\hat{y}(\emptyset) = 0$ and $\hat{y}(x) = b^\top X$, where $b \in \mathbb{R}^d$. For a fixed b, we define the following scalar fields:

$$F(x, u) := (2u - b^\top x)b^\top x, \tag{6}$$

$$G(x, y) := (2y - b^\top x)b^\top x. \tag{7}$$

The Sender's best response action is then characterized by the following lemma.

Lemma 2. *In the voluntary disclosure setup, for a given receiver strategy $\hat{y}(M)$, the Sender's best response action is given by*

$$A(x,u) = x \iff F(x,u) > 0. \tag{8}$$

Based on this best response, we derive the Receiver's strategic risk R_S in the next lemma.

Lemma 3. *Given a Receiver strategy $\hat{y}(M)$ parameterized by b, the resulting strategic risk is*

$$R_S(b,q) = \mathrm{Var}(Y) - (1-q)\mathbb{E}[G(X,Y)\mathbb{I}\{F(X,U) > 0\}], \tag{9}$$

where $\mathbb{I}\{\cdot\}$ denotes the indicator function.

Remark 2. *The Stackelberg equilibrium risk is defined as the minimum strategic risk over the parameter space:*

$$R_S^\star(q) := \min_b R_S(b,q).$$

Remark 3. *The function $F(x,u)$ serves as the decision rule for the Sender. The function $G(x,y)$ represents the Receiver's gain when the feature vector x is successfully received through the noisy channel (i.e., the signal is not lost to noise) relative to when the feature vector is lost.*

The strategic advantage condition is $R_V^\star(q) - R_S^\star(q) > 0$. We provide a sufficiency condition for this inequality in the following theorem.

Theorem 1 (Sufficiency Condition)**.** *A strategic advantage exists if there exists $b \in \mathbb{R}^d$ such that:*

$$\begin{aligned}\|U-Y\|_2 < {} & \frac{\mathbb{E}\big[|G(X,Y)|\big] - \mathrm{Var}(b_Y^\top X)}{4\sqrt{\mathrm{Var}(b_Y^\top X)}} \\ & - \frac{\mathrm{Var}((b-b_Y)^\top X)}{4\sqrt{\mathrm{Var}(b_Y^\top X)}},\end{aligned} \tag{10}$$

where $\|U-Y\|_2 = \sqrt{\mathrm{Var}(U) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(U,Y)}$, $b_Y = \mathrm{Var}(X)^{-1}\,\mathrm{Cov}(X,Y)$, and $G(X,Y) = (2Y - b^\top X)b^\top X$.

Beyond sufficiency, we can also identify a necessary condition for a strategic advantage to be possible. We define a strategic disadvantage when $R_S^\star(q) = R_V^\star(q)$; therefore, a necessary condition for an advantage must be the complement of this state.

Theorem 2 (Necessary Condition)**.** *A necessary condition for a strategic advantage to exist is:*

$$\begin{aligned}\max_b \Big(\mathbb{E}\big[|G(X,Y)|\big] - \mathrm{Var}(b_Y^\top X) \\ - \mathrm{Var}((b-b_Y)^\top X)\Big) > 0.\end{aligned} \tag{11}$$

Remark 4. *The necessary condition in* (11) *is equivalent to the requirement that the sufficiency condition in* (10) *be non-vacuous. Since the left-hand side of* (10) *is non-negative, the condition in* (11) *is equivalent to the numerator on the right-hand side being positive (as the denominator is always positive), thereby allowing for the existence of a non-empty range for $\|U-Y\|_2$.*

Remark 5. *Both the necessary and sufficient conditions are independent of the channel noise parameter q. This highlights that channel noise only controls the magnitude of the strategic advantage, not whether it can be attained. Additionally, both theorems require the bounds to hold for the best possible parameter b. Consequently, if the conditions are satisfied evaluated simply at the vanilla optimal weight b_Y, they necessarily hold under optimal b.*

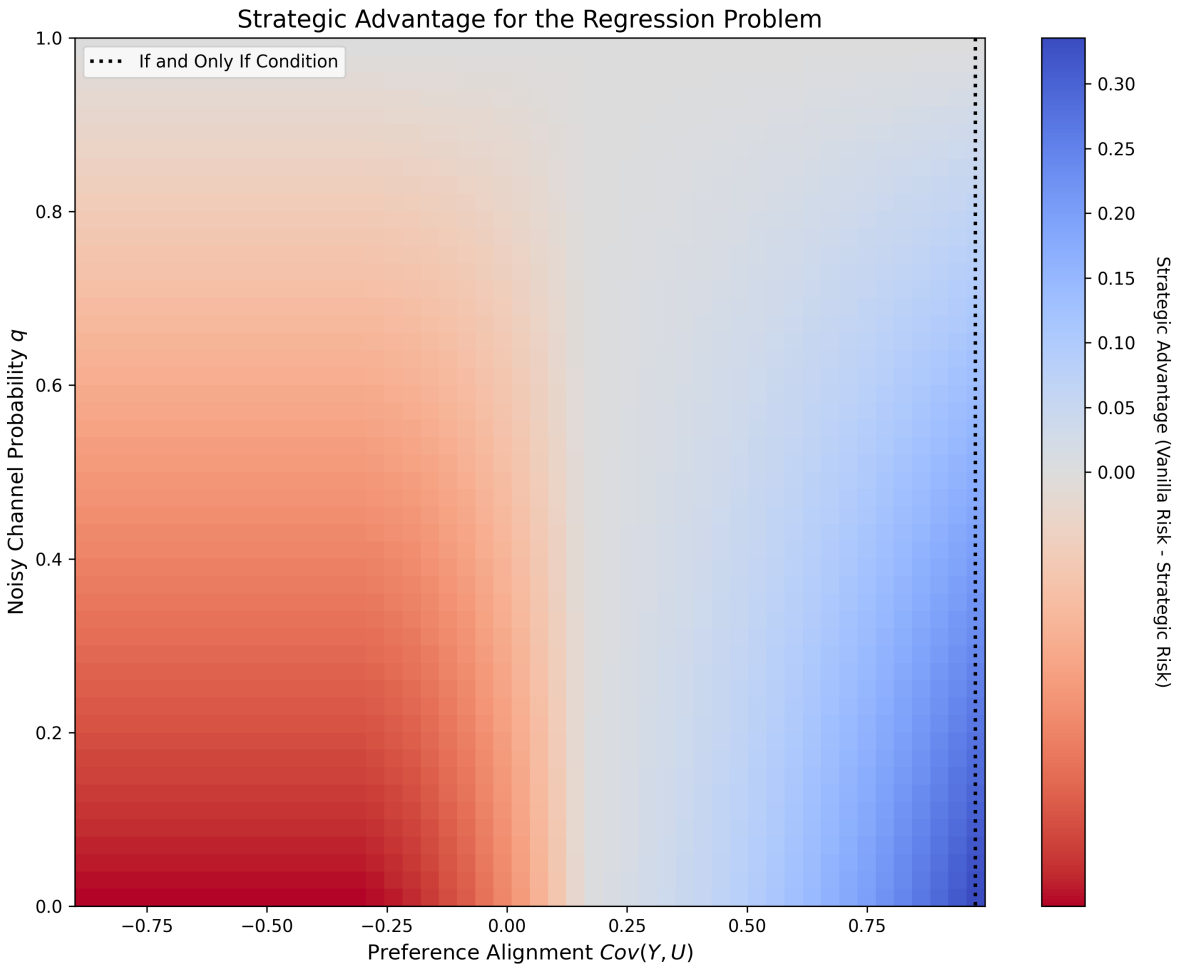

Figure 1: Heatmap of the strategic advantage $(R_V^\star(q) - R_S^\star(q) > 0)$ for the general regression setup. The parameter space sweeps the noisy channel probability q along the y-axis and the preference alignment $\mathrm{Cov}(Y, U)$ along the x-axis. The dotted black contour delineates the region where the sufficiency condition is met.

To numerically verify these theoretical results, we simulate a regression environment using a family of zero-mean Gaussian distributions for (X, Y, U) with unit marginal variances. We fix $\mathrm{Cov}(X, Y) = \mathrm{Cov}(X, U) = 0.2$ across this family, while $\mathrm{Cov}(Y, U)$ serves as the varying parameter. To explore the parameter space, we sweep the channel noise probability $q \in [0, 1]$ and the alignment $\mathrm{Cov}(Y, U) \in [-1, 1]$. At each point on the grid, the Receiver's risk is minimized over a range of weights b to calculate the strategic risk R_S. As demonstrated in Fig. 1, the analytical boundary reliably delineates the regions of strategic advantage.

3.2 Strategic Advantage Condition in Classification

We now sharpen our problem formulation (Section 2) for the classification setup and only mention the difference.

The outcome space $\mathcal{Y} = \{-1, +1\}$ is the space of discrete labels and the loss function $\ell(y, \hat{y}) = \mathbb{I}\{\hat{y} \neq y\}$ is the 0-1 loss. In the mandatory disclosure setup, the Sender always chooses $A = X$. For a Receiver strategy defined by $\hat{y}_V(\emptyset) = \mathrm{sign}(\mathbb{E}[Y])$ and $\hat{y}_V(x) = \mathrm{sign}(\mathbb{E}[Y|X = x])$, we characterize the vanilla resulting risk in the following lemma.

Lemma 4 (Classification Vanilla Risk)**.** *In the mandatory disclosure setup, the expected vanilla risk with noisy channel q is*

$$\begin{aligned} R_V^\star(q) &= q\mathbb{P}(Y = -\mathrm{sign}(\mathbb{E}[Y])) \\ &\quad + (1-q)\mathbb{E}\Big[\frac{1 - |\eta(X)|}{2}\Big], \end{aligned} \tag{12}$$

where $\eta(X) = \mathbb{E}[Y|X = x]$.

In the voluntary disclosure setup, the Receiver commits to a default prediction $\hat{y}(\emptyset) = i \in \{-1, +1\}$, which we refer to as the default option. A Sender with preference u observes (x, u) and chooses whether to disclose x. Under the 0-1 loss, the Sender withholds the feature if the default option matches their preference ($u = i$) and discloses it if $u = -i$ to potentially shift the prediction. If the Receiver observes the message $M = X$, the disclosure reveals that $u = -i$, prompting the Receiver to use the optimal posterior predictor $\hat{y}(x) = \mathrm{sign}(\mathbb{E}[Y \mid X = x, U = -i])$.

Because the optimum Receiver strategies are governed entirely by their choice of the default option i, there are only two possible strategies in the voluntary setup. Let us analyze each of them. Assuming without loss of generality that $\mathbb{E}[Y] > 0$, we denote the strategy where the default option matches the sign of the prior mean as $\mathcal{S}_1$ (i.e., $i = 1$) and the alternative strategy as $\mathcal{S}_{-1}$. We characterize the resulting strategic risk for both of these cases in the following lemma.

Lemma 5 (Strategic Classification Risk)**.** *For a default prediction $i \in \{-1, 1\}$, let $\pi_i = \mathbb{P}(U = i)$ and $\eta_{-i}(X) = \mathbb{E}[Y \mid X, U = -i]$. The strategic risk $R^\star_{\mathcal{S}_i}(q)$ is:*

$$\begin{aligned} R^\star_{\mathcal{S}_i}(q) &= \pi_i \mathbb{P}(Y = -i \mid U = i) \\ &\quad + \pi_{-i}\Bigg[(1-q)\mathbb{E}\left[\frac{1 - |\eta_{-i}(X)|}{2}\right] \\ &\quad + q\mathbb{P}(Y = -i \mid U = -i)\Bigg]. \end{aligned} \tag{13}$$

To determine the Stackelberg equilibrium, the Receiver must select the strategy $\mathcal{S}_i$ that has the minimum risk, i.e., $R^\star_S(q) = \min_i R^\star_{\mathcal{S}_i}(q)$. Since the channel noise parameter q acts linearly on the risk functions, we can characterize the optimal policy based on the value of q. By substituting the risk expressions from Lemmas 4 and 5, we formally characterize both the Stackelberg equilibrium and the conditions for strategic advantage.

Theorem 3 (Stackelberg Equilibrium and Strategic Advantage)**.** *Assume* $\mathbb{E}[Y] > 0$*. At the maximum noise limit (*$q = 1$*),* $\mathcal{S}_1$ *is strictly better (i.e.,* $R^\star_{\mathcal{S}_1}(1) < R^\star_{\mathcal{S}_{-1}}(1)$*). At* $q = 0$*, the optimal strategy depends on the underlying distributions, leading to two possible scenarios:*

1. *If* $R^\star_{\mathcal{S}_1}(0) \leq R^\star_{\mathcal{S}_{-1}}(0)$*, then* $\mathcal{S}_1$ *is the Stackelberg strategy for all* $q \in [0, 1]$*. The strategic advantage condition* $R^\star_V(q) > R^\star_S(q)$ *simplifies to:*

$$(1-q)\Bigg[\mathbb{E}\Big[\frac{1-|\eta(X)|}{2}\Big] - \pi_{-1}\mathbb{E}\Big[\frac{1-|\eta_{-1}(X)|}{2}\Big] - \pi_1\mathbb{P}(Y=-1 \mid U=1)\Bigg] > 0. \tag{14}$$

2. *If* $\mathcal{S}_{-1}$ *is better at* $q = 0$ *(i.e.,* $R^\star_{\mathcal{S}_1}(0) > R^\star_{\mathcal{S}_{-1}}(0)$*), there exists an intermediate threshold* $q^\star \in [0, 1]$ *given by*

$$q^\star = \frac{R^\star_{\mathcal{S}_1}(0) - R^\star_{\mathcal{S}_{-1}}(0)}{(R^\star_{\mathcal{S}_{-1}}(1) - R^\star_{\mathcal{S}_{-1}}(0)) - (R^\star_{\mathcal{S}_1}(1) - R^\star_{\mathcal{S}_1}(0))}, \tag{15}$$

such that for $q \geq q^\star$*,* $\mathcal{S}_1$ *is the Stackelberg strategy and the advantage condition is identical to* (14)*. For* $q < q^\star$*,* $\mathcal{S}_{-1}$ *is the Stackelberg strategy, and the strategic advantage requires:*

$$\begin{aligned}&(1-q)\Bigg[\mathbb{E}\Big[\frac{1-|\eta(X)|}{2}\Big] - \pi_1\mathbb{E}\Big[\frac{1-|\eta_1(X)|}{2}\Big]\Bigg]\\ &+ q\Bigg[\mathbb{P}(Y=-1) - \pi_1\mathbb{P}(Y=1 \mid U=1)\Bigg]\\ &> \pi_{-1}\mathbb{P}(Y=1 \mid U=-1).\end{aligned} \tag{16}$$

Remark 6. *In case (1) of Theorem 3, the condition in* (14) *is independent of* q*: the channel noise parameter only scales the magnitude of the advantage via the* $(1-q)$ *factor, analogous to the regression setting. In contrast, in case (2), the condition in* (16) *explicitly depends on* q*, meaning the channel noise governs both the magnitude and the direction of the strategic advantage.*

To empirically verify these theoretical results, we simulate the classification regime over a family of distributions. We assume discrete labels $(Y, U) \in \{-1, 1\}^2$ sampled from a joint distribution with fixed marginals $P(Y = 1) = 0.6$ and $P(U = 1) = 0.5$, while we vary $\mathrm{Cov}(Y, U)$ as the varying parameter. The features follow a Gaussian mixture model $X \mid Y, U \sim Y\mu_Y + U\mu_U + \mathcal{N}(0, 1)$, with $\mu_Y = 1.5$ and $\mu_U = 0.5$. To explore the parameter space, we sweep the channel noise probability $q \in [0, 1]$ and the alignment $\mathrm{Cov}(Y, U) \in [-1, 1]$. At each point on the grid, the strategic risk $R^\star_S(q)$ is evaluated by selecting the optimal Receiver strategy. As demonstrated in Fig. 2, the analytical boundary reliably delineates the regions of strategic advantage.

4 Online Learning for the regression problem

(**In Progress**)
In Section 3.1, we analyzed the optimal disclosure regimes under the assumption that the environment is known. We now extend the analysis to a dynamic setting in which the underlying environment parameters—specifically, the joint distribution of (X, Y, U)—are unknown to the Receiver. We focus on the setting where the Receiver must learn these parameters over time through repeated

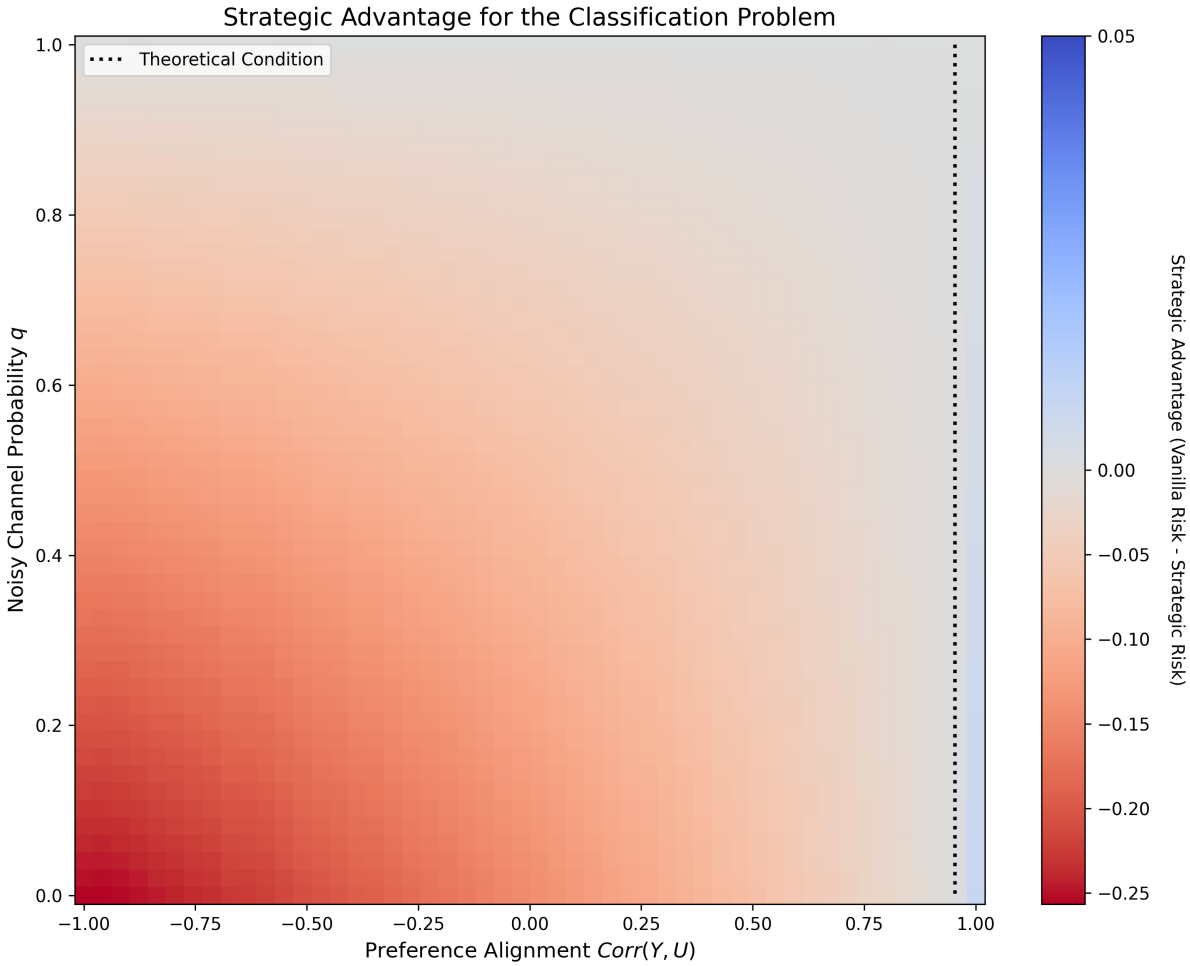

Figure 2: Heatmap of the strategic advantage $(R_V^\star(q) - R_S^\star(q) > 0)$ for the classification problem. The parameter space sweeps the noisy channel probability q along the y-axis and the discrete preference alignment $\mathrm{Corr}(Y, U)$ along the x-axis. The dotted black contour delineates the theoretical condition boundary from Theorem 3.

interaction with the Sender. Consider a sequence of T interactions. At each round $t \in [1, T]$, the protocol begins with the Receiver selecting a disclosure regime $D_t \in \{\text{Mandatory}, \text{Voluntary}\}$. Immediately following the regime selection, the Receiver commits to a predictor $\hat{y}_t$ tailored to the chosen regime. Subsequently, Nature draws the realization (x_t, u_t, y_t). The Sender, having observed the committed predictor $\hat{y}_t$, provides a best-response message M_t according to the rules of the selected regime D_t. Finally, the Receiver applies the predictor $\hat{y}_t(M_t)$ to the received message, incurs the loss $\ell(y_t, \hat{y}_t(M_t))$, and observes the ground truth outcome y_t. The Receiver's objective is to minimize the cumulative regret $\mathcal{R}_T$ over the horizon T, defined as

$$\mathcal{R}_T = \sum_{t=1}^{T} \ell\big(y_t, \hat{y}_t(M_t)\big) - T \cdot \min\{R_V^\star, R_S^\star\}, \tag{17}$$

where $R_V^\star$ and $R_S^\star$ denote the optimal voluntary and strategic risks that would be achieved if the environment were known.

To model uncertainty, suppose the environment is parameterized by $\theta \in \Theta$, where Θ denotes a family of admissible joint distributions. The Receiver maintains a belief density $p_t(\theta)$ over Θ at time t to balance exploration and exploitation. This interaction is formalized in Algorithm 1.

Algorithm 1 Bayesian Adaptive Disclosure Protocol

Initialize: Prior belief $p_1(\theta)$ over parameter space Θ.
for $t = 1$ to T **do**
  Evaluate risk: For regime $D \in \{\text{Mandatory}, \text{Voluntary}\}$ and predictor $\hat{y}$, compute expected loss under $p_t(\theta)$.
  Selection: Choose $(D_t, \hat{y}_t)$ minimizing the expected loss to balance exploration and exploitation.
  Commitment: Commit to predictor $\hat{y}_t$ under regime D_t.
  Interaction: Sender observes $\hat{y}_t$ and provides best-response message M_t.
  Realization: Ground truth y_t is revealed; Receiver incurs loss $\ell(y_t, \hat{y}_t(M_t))$.
  Belief Update: Update posterior using Bayes' rule:

$$p_{t+1}(\theta) \propto p_t(\theta) \cdot \mathbb{P}(M_t, y_t \mid \theta, \hat{y}_t)$$

end for

The following theorem analyses the regret of this algoritm.

Theorem 4 (Regret of Adaptive Disclosure)**.** *Assume that Θ is compact and that the likelihood functions satisfy standard regularity conditions. Then the Bayesian adaptive disclosure policy described in Algorithm 1 achieves sublinear expected regret:*

$$\mathbb{E}[\mathcal{R}_T] \leq C\sqrt{T}, \tag{18}$$

where C depends on the statistical complexity and information-theoretic properties of the environment class.

5 CONCLUSION

In this paper, we have provided formal sufficiency and necessity conditions for strategic advantages in both regression and classification settings. These conditions highlight the role of misalignment and the noisy channel parameter q in the receiver's choice between forced and optional revelation modes. Future work includes the extension to multi-sender environments.

6 POSTPONED PROOFS

Here we present the proof of all results we skipped in the Section 3.

6.1 Proofs for Strategic Advantage Condition for Regression

This section provides the complete derivations for the general regression risk formulations and the subsequent sufficiency and necessity conditions.

Proof of Lemma 1. We compute the risk in the mandatory disclosure setting for a linear predictor $b \in \mathbb{R}^d$ with $\hat{y}_V(\emptyset) = 0$ and $\hat{y}_V(x) = b^\top x$. The risk is given by:

$$R_V(b, q) = q\mathbb{E}[Y^2] + (1-q)\mathbb{E}[(Y - b^\top X)^2].$$

Since all variables have mean 0, $\mathrm{Var}(Y) = \mathbb{E}[Y^2]$. We now examine the term $\mathbb{E}[(Y - b^\top X)^2]$. We add and subtract $b_Y^\top X$ inside the expectation, where $b_Y = \mathrm{Var}(X)^{-1}\,\mathrm{Cov}(X, Y)$:

$$\begin{aligned}\mathbb{E}[(Y - b^\top X)^2] = \mathbb{E}[(Y - b_Y^\top X)^2] \\ + \mathbb{E}[((b_Y - b)^\top X)^2] \\ + 2\mathbb{E}[(Y - b_Y^\top X)(b_Y - b)^\top X].\end{aligned}$$

By the orthogonality principle of OLS, $\mathbb{E}[(Y - b_Y^\top X)X] = 0$, so the cross-term vanishes. Thus,

$$\mathbb{E}[(Y - b^\top X)^2] = \mathrm{Var}(Y) - \mathrm{Var}(b_Y^\top X) + \mathrm{Var}((b - b_Y)^\top X).$$

Substituting back into the risk yields (4). The optimal strategy minimizes $\mathrm{Var}((b - b_Y)^\top X)$ by choosing $b = b_Y$. □

Proof of Lemma 2. Given a realization (x, u), the Sender selects the action $A \in \{\emptyset, x\}$ minimizing $(u - \hat{y}(M))^2$. The expected loss from revealing is $q\,u^2 + (1-q)(u - b^\top x)^2$, while from withholding it is u^2. The Sender prefers revealing if:

$$u^2 > q\,u^2 + (1-q)(u - b^\top x)^2.$$

Simplifying yields $(b^\top x)(2u - b^\top x) > 0$, which is $F(x, u) > 0$. □

Proof of Lemma 3. For a fixed parameter b, the strategy of the Receiver is $\hat{y}_V(\emptyset) = 0$ for an empty message $M = \emptyset$ and $\hat{y}_V(x) = b^\top x$ for a successfully received feature vector $M = X$.

According to Lemma 2, the Sender reveals the feature vector x if and only if $F(X, U) > 0$. Using the law of total expectation, we decompose the strategic risk $R_S(b, q) = \mathbb{E}[\ell(Y, \hat{y}(M))]$ into the two cases of F:

$$\begin{aligned}R_S(b, q) &= \mathbb{E}\big[\mathbb{I}\{F \le 0\}\ell(Y, \hat{y}(\emptyset)) \\ &\quad + \mathbb{I}\{F > 0\}(q\ell(Y, \hat{y}(\emptyset)) + (1-q)\ell(Y, \hat{y}(X)))\big] \\ &= \mathbb{E}\big[\mathbb{I}\{F \le 0\}Y^2 \\ &\quad + \mathbb{I}\{F > 0\}(qY^2 + (1-q)(Y - b^\top X)^2)\big].\end{aligned}$$

Applying the identity $\mathbb{I}\{F \le 0\} = 1 - \mathbb{I}\{F > 0\}$:

$$\begin{aligned}R_S(b, q) &= \mathbb{E}[Y^2] - (1-q)\mathbb{E}\big[\mathbb{I}\{F > 0\}(Y^2 - (Y - b^\top X)^2)\big] \\ &= \mathrm{Var}(Y) - (1-q)\mathbb{E}\big[\mathbb{I}\{F > 0\}(2Y(b^\top X) \\ &\quad - (b^\top X)^2)\big],\end{aligned}$$

where $\mathrm{Var}(Y) = \mathbb{E}[Y^2]$ follows from the zero-mean assumption. Substituting $G(X,Y) = (2Y - b^\top X)b^\top X$ yields:

$$R_S(b,q) = \mathrm{Var}(Y) - (1-q)\mathbb{E}[\mathbb{I}\{F > 0\}G(X,Y)],$$

which concludes the proof. □

Proof of Theorem 1 (Sufficiency Condition). We seek condition such that $R_S^\star(q) - R_V^\star(q) < 0$. From Lemmas 1 and 3:

$$\begin{aligned} R_S^\star(q) - R_V^\star(q) = \mathrm{Var}(Y) &- (1-q)\mathbb{E}[G\mathbb{I}\{F > 0\}] \\ &- \big(\mathrm{Var}(Y) - (1-q)\,\mathrm{Var}(b_Y^\top X)\big). \end{aligned}$$

Using $\mathbb{I}\{F > 0\} = 1 - \mathbb{I}\{F \le 0\}$, we have $\mathbb{E}[G\mathbb{I}\{F > 0\}] = \mathbb{E}[G] - \mathbb{E}[G\mathbb{I}\{F \le 0\}]$.

To compute $\mathbb{E}[G]$, recall $G(X,Y) = (2Y - b^\top X)b^\top X$. Since $\mathbb{E}[X] = 0$, $\mathrm{Var}(b^\top X) = \mathbb{E}[(b^\top X)^2]$. Furthermore, the optimal OLS weights $b_Y = \mathbb{E}[XX^\top]^{-1}\mathbb{E}[XY]$ satisfy $\mathbb{E}[Y(b^\top X)] = b^\top \mathbb{E}[XY] = b^\top \mathrm{Var}(X) b_Y = \mathrm{Cov}(b^\top X, b_Y^\top X)$. Thus,

$$\begin{aligned} \mathbb{E}[G] &= 2\mathbb{E}[Y(b^\top X)] - \mathbb{E}[(b^\top X)^2] \\ &= 2b^\top \mathrm{Var}(X) b_Y - \mathrm{Var}(b^\top X). \end{aligned}$$

From the identity $\mathrm{Var}((b - b_Y)^\top X) = \mathrm{Var}(b^\top X) + \mathrm{Var}(b_Y^\top X) - 2b^\top \mathrm{Var}(X) b_Y$, we obtain:

$$\mathbb{E}[G] = \mathrm{Var}(b_Y^\top X) - \mathrm{Var}((b - b_Y)^\top X).$$

Substituting this into the risk difference we get:

$$R_S^\star(q) - R_V^\star(q) = (1-q)\big(\mathrm{Var}((b - b_Y)^\top X) + \mathbb{E}[G\mathbb{I}\{F \le 0\}]\big).$$

We upper bound $\mathbb{E}[G\mathbb{I}\{F \le 0\}]$ by decomposing:

$$\begin{aligned} \mathbb{E}[G\mathbb{I}\{F \le 0\}] &= \mathbb{E}[G\mathbb{I}\{G \le 0\}] \\ &\quad + \mathbb{E}[G(\mathbb{I}\{F \le 0\} - \mathbb{I}\{G \le 0\})] \\ &= -\tfrac{1}{2}\big(\mathbb{E}[|G|] - \mathbb{E}[G]\big) \\ &\quad + \mathbb{E}[G\mathbb{I}\{\mathrm{sign}(F) \ne \mathrm{sign}(G)\}]. \end{aligned}$$

When $\mathrm{sign}(F) \ne \mathrm{sign}(G)$, we have $|G| \le |G - F| = 2|Y - U||b^\top X|$. By Cauchy-Schwarz:

$$\begin{aligned} \mathbb{E}[G\mathbb{I}\{\mathrm{sign}(F) \ne \mathrm{sign}(G)\}] &\le \mathbb{E}[2|Y - U|\,|b^\top X|] \\ &\le 2\|U - Y\|_2 \sqrt{\mathrm{Var}(b_Y^\top X)}. \end{aligned}$$

Setting upper bound to be negative, (which would imply $R_S^\star(q) - R_V^\star(q) < 0$), we get:

$$\begin{aligned} &\mathrm{Var}((b - b_Y)^\top X) \\ &- \tfrac{1}{2}\big(\mathbb{E}[|G|] - \mathrm{Var}(b_Y^\top X) \\ &\quad + \mathrm{Var}((b - b_Y)^\top X)\big) \\ &+ 2\|U - Y\|_2 \sqrt{\mathrm{Var}(b_Y^\top X)} < 0. \end{aligned}$$

Rearranging isolates $\|U - Y\|_2$, yielding (10). □

Proof of Theorem 2 (Necessary Condition). We prove the contrapositive: if (11) fails, then no strategic advantage can exist, i.e., $R_S(b,q) \geq R_V^\star(q)$ for all b. From Lemma 1 and Lemma 3, we have:

$$R_S(b,q) - R_V^\star(q) = (1-q)\big(\operatorname{Var}((b-b_Y)^\top X) + \mathbb{E}[G\mathbb{I}\{F \leq 0\}]\big).$$

A strategic advantage requires $R_S(b,q) - R_V^\star(q) < 0$ for some b. We can lower bound the term $\mathbb{E}[G\mathbb{I}\{F \leq 0\}]$ purely algebraically.

If $G \leq 0$, we have $G\mathbb{I}\{F \leq 0\} \geq G = G\mathbb{I}\{G \leq 0\}$. If $G > 0$, we have $G\mathbb{I}\{F \leq 0\} \geq 0 = G\mathbb{I}\{G \leq 0\}$. Therefore, pointwise $G\mathbb{I}\{F \leq 0\} \geq G\mathbb{I}\{G \leq 0\}$, which implies:

$$\mathbb{E}[G\mathbb{I}\{F \leq 0\}] \geq \mathbb{E}[G\mathbb{I}\{G \leq 0\}].$$

Substituting this inequality yields:

$$R_S(b,q) - R_V^\star(q) \geq (1-q)\big(\operatorname{Var}((b-b_Y)^\top X) + \mathbb{E}[G\mathbb{I}\{G \leq 0\}]\big).$$

Now $\mathbb{E}[G\mathbb{I}\{G \leq 0\}]$ depends only on (X,Y) and b, with U entirely eliminated. Using the identity $\mathbb{E}[G\mathbb{I}\{G \leq 0\}] = -\frac{1}{2}(\mathbb{E}[|G|] - \mathbb{E}[G])$ and $\mathbb{E}[G] = \operatorname{Var}(b_Y^\top X) - \operatorname{Var}((b-b_Y)^\top X)$:

$$\begin{aligned} R_S(b,q) - R_V^\star(q) \geq (1-q)\Big(&\operatorname{Var}((b-b_Y)^\top X) \\ &- \tfrac{1}{2}\big(\mathbb{E}[|G|] - \operatorname{Var}(b_Y^\top X) \\ &+ \operatorname{Var}((b-b_Y)^\top X)\big)\Big). \end{aligned}$$

Multiplying by 2 and rearranging, $R_S(b,q) - R_V^\star(q) < 0$ requires:

$$\mathbb{E}[|G|] - \operatorname{Var}(b_Y^\top X) - \operatorname{Var}((b-b_Y)^\top X) > 0.$$

If this fails for all b, then $R_S(b,q) - R_V^\star(q) \geq 0$ for all b, meaning no strategic advantage is possible. Hence the necessary condition is that there exists some b satisfying (11) and thus

$$\max_b \mathbb{E}[|G|] - \operatorname{Var}(b_Y^\top X) - \operatorname{Var}((b-b_Y)^\top X) > 0.$$

□

6.2 Proofs for Strategic Advantage Condition for Classification

Proof of Lemma 4. Under the mandatory disclosure setup (vanilla benchmark), the Receiver implements $\hat{y}_V(\emptyset) = \operatorname{sign}(\mathbb{E}[Y])$ when $M = \emptyset$, and $\hat{y}_V(x) = \operatorname{sign}(\mathbb{E}[Y \mid X = x])$ when $M = X$. Let $\eta(x) = \mathbb{E}[Y \mid X = x]$ and $p(x) = \mathbb{P}(Y = 1 \mid X = x)$. Since $Y \in \{-1, +1\}$, we have $\eta(x) = 2p(x) - 1$. When $\eta(x) \geq 0$, the optimum prediction is $+1$, yielding an error probability of $1 - p(x)$. When $\eta(x) < 0$, the optimum prediction is -1, yielding an error probability of $p(x)$. In both cases, the conditional Bayes risk is equivalently given by $(1 - |\eta(x)|)/2$. Thus, the overall expected risk when $M = X$ is $\mathbb{E}[(1 - |\eta(X)|)/2]$.

When $M = \emptyset$, the constant classifier yields risk $\mathbb{P}(Y \neq \operatorname{sign}(\mathbb{E}[Y]))$, which is equal to $\mathbb{P}(Y = -\operatorname{sign}(\mathbb{E}[Y]))$. Combining these mutually exclusive events with their respective probabilities q and $1 - q$ under the noisy channel yields the stated vanilla risk in (12). □

Proof of Lemma 5. For a chosen default prediction $\hat{y}(M) = i$ when $M = \emptyset$, the Receiver correctly learns from a non-empty message $M = X$ that $U = -i$. Consequently, the optimal conditional strategy becomes $\hat{y}_V(x) = \text{sign}(\mathbb{E}[Y \mid X = x, U = -i])$. If the Sender has preference $U = i$, withholding features gives a loss of 0 (since the Receiver plays i), while revealing may risk a different prediction. Thus, the Sender strictly withholds the document, causing $M = \emptyset$. The conditional risk for this group ($U = i$) is simply the risk of the default prediction: $\mathbb{P}(Y = -i \mid U = i)$. If the Sender has preference $U = -i$, withholding yields the worst possible loss for them. Hence, they choose to reveal $A = X$. The message drops with probability q (leading to prediction i and contributing $q\mathbb{P}(Y = -i \mid U = -i)$ to the error), and successfully arrives with probability $1 - q$, where the conditional Bayes risk over the subgroup is $\mathbb{E}[(1 - |\eta_{-i}(X)|)/2]$. Multiplying the expectations by their respective cluster probabilities $\pi_i = \mathbb{P}(U = i)$ exactly formulates the overall strategic risk of $\mathcal{S}_i$ as (13). □

Proof of Theorem 3. The risk functions $R^\star_{\mathcal{S}_1}(q)$ and $R^\star_{\mathcal{S}_{-1}}(q)$ from Lemma 5 are linear equations in q. Assuming $\mathbb{E}[Y] > 0$, which implies $\mathbb{P}(Y = 1) > \mathbb{P}(Y = -1)$. At maximum channel noise ($q = 1$), the messages are completely dropped ($M = \emptyset$), reducing the Receiver risk to the static default prediction's baseline error. Hence, $R^\star_{\mathcal{S}_1}(1) = \mathbb{P}(Y = -1)$, and $R^\star_{\mathcal{S}_{-1}}(1) = \mathbb{P}(Y = 1)$. Since $\mathbb{P}(Y = 1) > \mathbb{P}(Y = -1)$, it is clear that $R^\star_{\mathcal{S}_1}(1) < R^\star_{\mathcal{S}_{-1}}(1)$, making $\mathcal{S}_1$ strictly optimal at $q = 1$. Because both risks are lines as functions of q, if $R^\star_{\mathcal{S}_1}(0) \leq R^\star_{\mathcal{S}_{-1}}(0)$ (lets call it scenarion A), $\mathcal{S}_1$ is optimal at both extremes and thus dominates everywhere via linear interpolation. Conversely, if $R^\star_{\mathcal{S}_{-1}}(0) < R^\star_{\mathcal{S}_1}(0)$ (lets call it scenario B), the linear functions cross once inside the interval $q \in (0, 1)$. This intersection naturally defines the transition threshold $q^\star$ given by

$$q^\star = \frac{R^\star_{\mathcal{S}_1}(0) - R^\star_{\mathcal{S}_{-1}}(0)}{(R^\star_{\mathcal{S}_{-1}}(1) - R^\star_{\mathcal{S}_{-1}}(0)) - (R^\star_{\mathcal{S}_1}(1) - R^\star_{\mathcal{S}_1}(0))}, \tag{19}$$

such that for $q < q^\star$, $\mathcal{S}_{-1}$ is optimal, and for $q \geq q^\star$, $\mathcal{S}_1$ is optimal. We obtain the strategic advantage condition by verifying when $R^\star_V(q) - R^\star_{\mathcal{S}_i}(q) > 0$ for the optimal i. Under the domain where $\mathcal{S}_1$ is Stackelberg-optimal (all q in scenario A, or $q \geq q^\star$ in scenario B):

$$R^\star_V(q) - R^\star_{\mathcal{S}_1}(q) > 0.$$

Using $\text{sign}(\mathbb{E}[Y]) = 1$, we decompose the baseline error rate in $R^\star_V(q)$:

$$\begin{aligned} q\mathbb{P}(Y = -1) = q[&\pi_1 \mathbb{P}(Y = -1 \mid U = 1) \\ &+ \pi_{-1}\mathbb{P}(Y = -1 \mid U = -1)]. \end{aligned}$$

Upon substituting $R^\star_V(q)$ and $R^\star_{\mathcal{S}_1}(q)$, the terms matching $q\pi_{-1}\mathbb{P}(Y = -1 \mid U = -1)$ cancel algebraically. The remaining terms structurally share a common $(1 - q)$ factor. Dividing the strict inequality by $(1 - q)$ yields the noise-invariant condition shown in (14). In the alternate domain where $\mathcal{S}_{-1}$ is strictly optimal ($q < q^\star$ in scenario B), we calculate $R^\star_V(q) - R^\star_{\mathcal{S}_{-1}}(q) > 0$. Here, the q dependencies no longer cancel out purely because the vanilla baseline ($+1$) and the strategic default choice (-1) are misaligned. Grouping the independent expectations corresponding to q and $(1 - q)$ distinctly and isolating terms directly formulates the threshold specified in (16). □

6.3 Online Learning Analysis

Proof of Theorem 4 (Outline). We decompose the cumulative regret into two components and bound each separately.

Step 1: Regret Decomposition. Let $D^\star \in \{\text{Mandatory}, \text{Voluntary}\}$ denote the optimal disclosure regime and $\hat{y}^\star$ the associated optimal predictor under the true environment $\theta^\star$. The instantaneous regret at round t is:

$$r_t = \ell(y_t, \hat{y}_t(M_t)) - \ell(y_t, \hat{y}^\star(M_t^\star)).$$

We decompose this as:

$$\begin{aligned} r_t = &\underbrace{\ell(y_t, \hat{y}_t(M_t)) - \ell(y_t, \hat{y}_t^\star(M_t))}_{\text{(I): estimation regret}} \\ &+ \underbrace{\ell(y_t, \hat{y}_t^\star(M_t)) - \ell(y_t, \hat{y}^\star(M_t^\star))}_{\text{(II): regime selection regret}}, \end{aligned}$$

where $\hat{y}_t^\star$ denotes the oracle-optimal predictor for the regime D_t chosen at round t.

Step 2: Bounding Regime Selection Regret (II). Term (II) is nonzero only when the algorithm selects a suboptimal regime $D_t \neq D^\star$. Define the suboptimality gap $\Delta = |R^\star_{D^\star} - R^\star_{D^c}|$ where D^c is the suboptimal regime. Under the Bayesian posterior update, posterior concentration yields:

$$\mathbb{P}(D_t \neq D^\star) \leq \mathbb{P}\big(\|\hat{\theta}_t - \theta^\star\| > \delta\big),$$

for an appropriate δ depending on Δ. By standard Bayesian posterior contraction under the compactness and regularity assumptions on Θ, the posterior concentrates at rate $1/\sqrt{t}$:

$$\mathbb{P}\big(\|\hat{\theta}_t - \theta^\star\| > \delta\big) \leq C_1 \exp\big(-c_1 t \delta^2\big).$$

Summing over t, the total contribution of regime misidentification rounds is:

$$\sum_{t=1}^{T} \mathbb{E}[(\text{II})_t] \leq C_2 \log T.$$

Step 3: Bounding Estimation Regret (I). Conditional on selecting the correct regime $D_t = D^\star$, term (I) reduces to standard online prediction regret. Under the mandatory regime, this is online linear regression against an i.i.d. sequence, which satisfies $\sum_t \mathbb{E}[(\text{I})_t \mid D_t = D^\star] \leq C_3\sqrt{T}$ via standard results (e.g., Vovk–Azoury–Warmuth bounds for squared loss).

Under the voluntary regime, the Receiver additionally needs to estimate the Sender's best-response mapping. Since the Sender's action depends on the committed predictor and the joint distribution, the effective observation model remains parametric. The Bayesian belief update in Algorithm 1 incorporates the likelihood $\mathbb{P}(M_t, y_t \mid \theta, \hat{y}_t)$, which accounts for the Sender's strategic behavior. The posterior mean predictor then achieves:

$$\sum_{t=1}^{T} \mathbb{E}[(\text{I})_t] \leq C_3\sqrt{T}.$$

Step 4: Combining. Adding both terms:

$$\mathbb{E}[\mathcal{R}_T] = \sum_{t=1}^{T} \mathbb{E}[r_t] \leq C_2 \log T + C_3\sqrt{T} \leq C\sqrt{T},$$

where the $\log T$ regime identification cost is absorbed into the dominant $\sqrt{T}$ estimation term. □

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